DATA – 211 Introduction to Data Science

BASIC STATISTICS

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TABLE

Faculty Name	Area of Specialization	Experience in yrs	International Faculty	# Assignments
Ahmad	Data Science	23	Yes	Medium
Ameen	Operations Research	10	No	High
Aqib	Simulation	13	Yes	High
Bashir	Data Science	20	No	Medium
Nadhir	Operations Research	18	Yes	Low

1.Numerical Data:

- These are numerical values
- For example, in the previous table, *Experience in yrs* is a numerical data. A value of 10 in the second row implies the corresponding faculty has 10 years of experience.

2. Categorical Data:

- there are non-numerical field values, where the value represents a category.
- For example, in the previous table, *International Faculty* is a categorical data. A value of 'Yes' in the first row implies the corresponding faculty a foreign faculty.

Categorical Data:

1. Nominal Data:

- We can use categorical data for convenience instead of numerical.
- For example, in *International Faculty* column, one could replace 'Yes' with 1 and 'No' with 0, or vice-versa. These numbers do not have meaningful mathematical or statistical insights. They are just representation for convenience.

2. Ordinal Data:

- The value represents a category with order.
- For example, in the previous table, *# Assignments* is a ordinal data. A value of 'High' in the second row implies the corresponding faculty assigns too much homework.

Field Segregation

Not all the fields in a given data table are same. They can be classified or segregated as:

- **Independent Variables:** A variable that is thought to be controlled or not affected by other variables is called an independent variable. Typically, it is also known as **attribute, control, explanatory, regressor, input, predictor, observed, feature, field** variable.
- **Dependent Variables:** A variable that depends on other variable (and maybe of interest in the analysis). It is also known as **predicted, explained, target, response, output/outcome, label** variable.
- **Auxiliary Variables:** These are the variables that provide meta information about the data and may not be helpful (or may not be meant) to be used in the analysis.

Stu ID	Score	Subj	GPA	Honor	Grade
2508	75	Math	2.97	0	B+
2679	81	Phys	3.25	0	B+
2416	95	Math	3.55	1	A
2720	85	Chem	3.12	0	B+
2575	81	Phys	3.09	0	B
2118	82	Math	3.33	0	B+
2060	97	Phys	3.78	1	A+

1. Identify all the variable/fields.
   Stu ID, Score, Subj, GPA, Honor, and Grade.

2. Identify the variable/field types. The following table gives the required information:

Field	Type
Stu ID	Nominal
Score	Numeric
Subj	Categorical
GPA	Numeric
Honor	Nominal
Grade	Ordinal

Histogram

It helps visualize the **distribution (shape) of the data**.

A **graphical method** used for **univariate data** (one variable at a time).

The data is divided into **classes (intervals or bins)**.

For each class, we count how many data points fall inside → this is the **frequency**.

We draw **rectangles (bars)** where:

x-axis = classes (data intervals)

y-axis = frequency (count)

Histogram Analysis

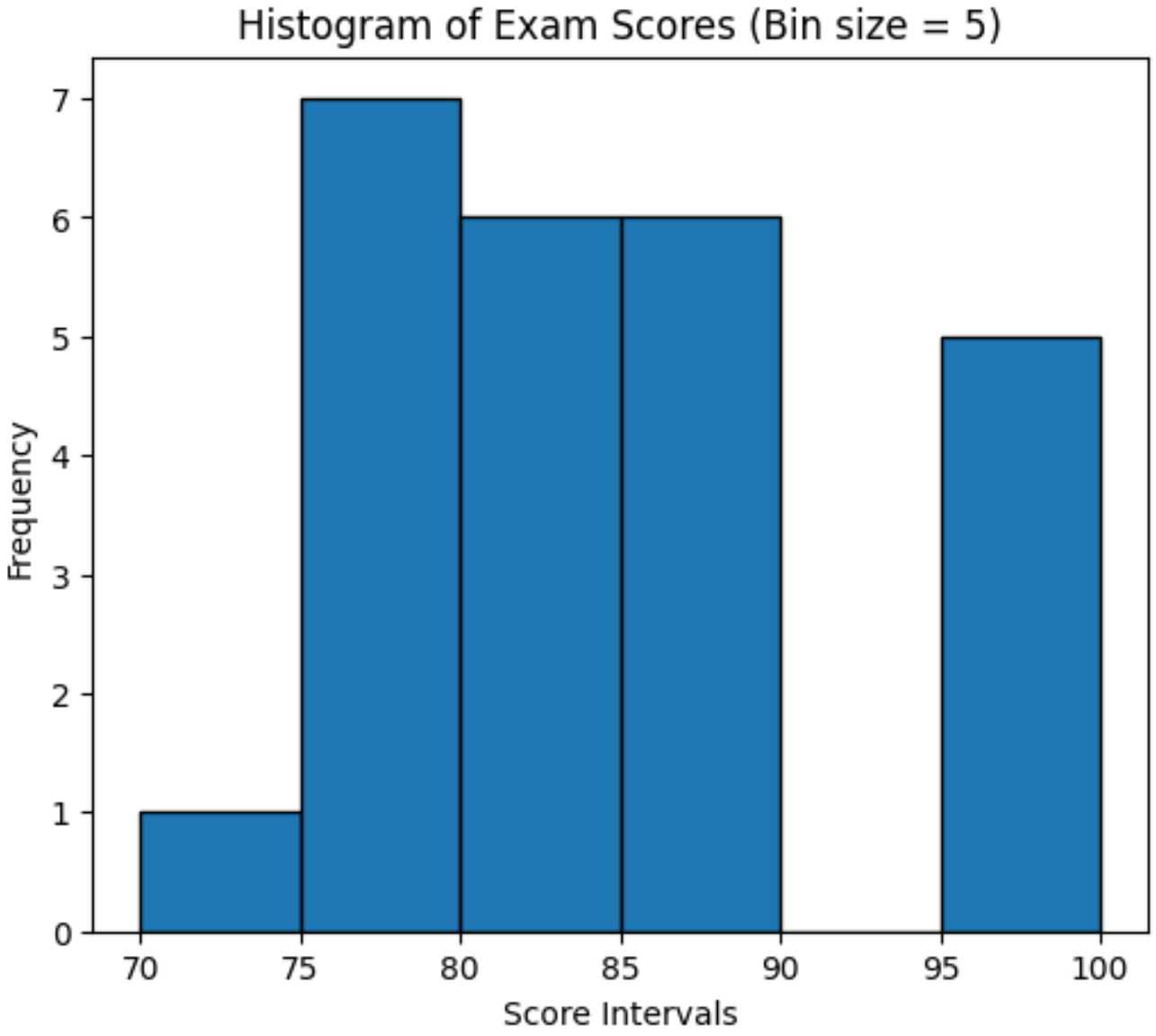
Exam scores = [71, 75, 75, 76, 77, 78, 79, 79, 81, 81, 81, 82, 82, 83, 85, 86, 87, 87, 87, 89, 95, 97, 98, 100, 100]

Choose class intervals (bins)

Suppose size =5

Interval	Scores inside	Frequency
70–74	71	1
75–79	75, 75, 76, 77, 78, 79, 79	7
80–84	81, 81, 81, 82, 82, 83	6
85–89	85, 86, 87, 87, 87, 89	6
90–94	–	0
95–100	95, 97, 98, 100, 100	5

Histogram Analysis

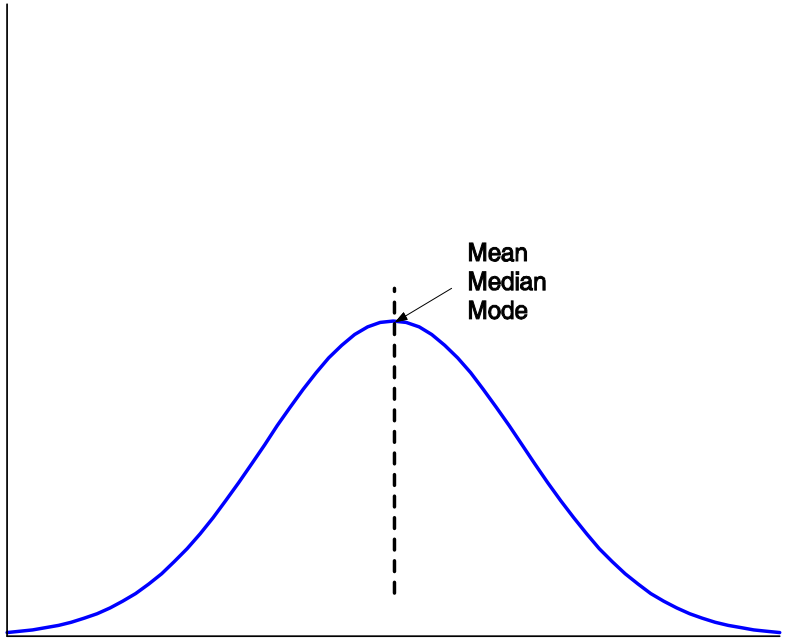


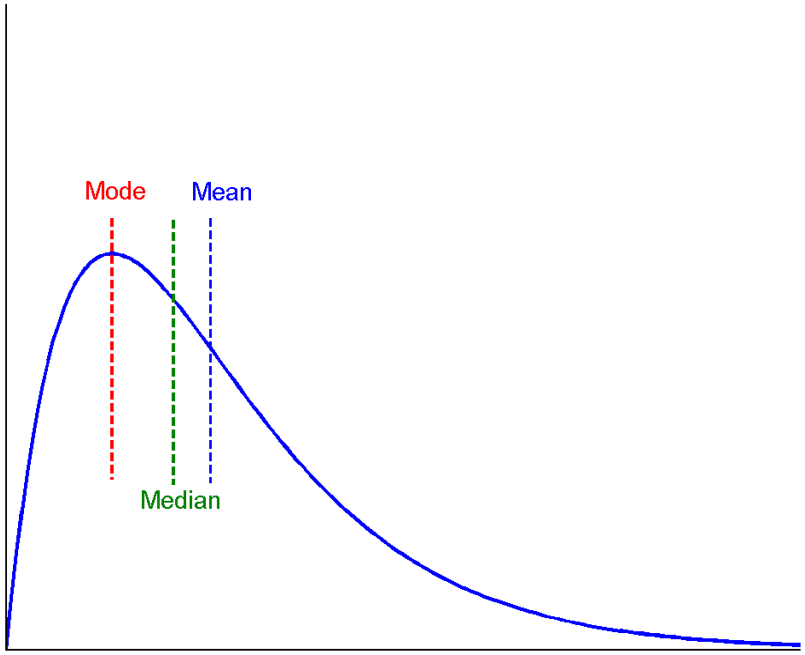
Measuring the Central Tendency

- Mean (algebraic measure) (sample vs. population):
  - Weighted arithmetic mean:
  - Trimmed mean: chopping extreme values
  - Example: (75+81+95)/3 = 83.67
- Median: A holistic measure
  - Middle value if odd number of values, or average of the middle two values otherwise
  - Example: 71, 75, 79 → Median = 75
- Mode
  - Value that occurs most frequently in the data
  - Unimodal, bimodal, trimodal
  - [81, 87, 87, 95] → Mode = 87
  - [81, 87, 87, 81, 95] → Mode = 81 & 87

Symmetric vs. Skewed Data

Median, mean and mode of symmetric, positively and negatively skewed data





Measuring the Dispersion of Data

The spread or dispersion of a distribution can be measured using the following typical measures: **Range, Interquartile Range, Variance** and **Standard Deviation**.

Scores = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 79, 87, 87, 79, 75, 100, 82, 83, 98, 86, 87, 71, 78, 81]

- **Range**

$$\text{Range} = Max - Min$$
$$= 100 - 71 = 29$$

Measuring the Dispersion of Data

Variance

Variance tells us **how spread out the numbers are** around the mean.

- If variance is **small**, numbers are close to the mean.
- If variance is **large**, numbers are more spread out.

The variance of the population is defined by the following formula:

$$\sigma^2 = \frac{\sum(X_i - \overline{X})^2}{N}$$

where σ^2 is the population variance, $\overline{X}$ is the population mean, X_i is the ith element from the population, and N is the number of elements in the population.
The variance of the sample is defined by the following formula:

$$s^2 = \frac{\sum(x_i - \overline{x})^2}{n - 1}$$

where s^2 is the sample variance, $\overline{x}$ is the sample mean, x_i is the ith element from the sample, and n is the number of elements in the sample.
The above formula, the variance of the sample is an unbiased estimate of the variance of the population.

Measuring the Dispersion of Data

Variance

Example Data

Suppose we have the following exam scores of 5 students:

$$2,\ 4,\ 4,\ 4,\ 6$$

Step 1: Find the Mean

$$\bar{x} = \frac{2+4+4+4+6}{5} = \frac{20}{5} = 4$$

So, the **mean = 4**.

Step 2: Find Deviations from the Mean

Subtract the mean (4) from each value:

$$(2-4) = -2,\ \ (4-4) = 0,\ \ (4-4) = 0,\ \ (4-4) = 0,\ \ (6-4) = 2$$

So deviations are: **-2, 0, 0, 0, 2**

Measuring the Dispersion of Data

Variance

Step 3: Square the Deviations

$$(-2)^2 = 4, \quad 0^2 = 0, \quad 0^2 = 0, \quad 0^2 = 0, \quad (2)^2 = 4$$

Squared deviations: **4, 0, 0, 0, 4**

Step 4: Average the Squared Deviations

- If this is the **population variance (σ²)**:

$$\sigma^2 = \frac{4 + 0 + 0 + 0 + 4}{5} = \frac{8}{5} = 1.6$$

- If this is a **sample variance (s²)**:

$$s^2 = \frac{4 + 0 + 0 + 0 + 4}{5 - 1} = \frac{8}{4} = 2$$

Measuring the Dispersion of Data

Scores = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 79, 87, 87, 79, 75, 100, 82, 83, 98, 86, 87, 71, 78, 81]

Number of values:

$$n = 25$$

Step 2: Mean

$$\bar{x} = \frac{\sum x_i}{n} = \frac{2111}{25} = 84.44$$

Step 3: Squared deviations

$$(x_i - \bar{x})^2 \quad \text{for each score}$$

When summed:

$$\sum (x_i - \bar{x})^2 = 1616.16$$

Step 4: Variance formulas

- **Population Variance (σ²):**

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{1616.16}{25} = 64.65$$

- **Sample Variance (s²):**

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{1616.16}{24} = 67.34$$

Measuring the Dispersion of Data

Standard Deviation (SD)

- Standard deviation is the **square root of variance**.
- It tells us **how much the data values spread out from the mean** in the **same units as the data**

Population Standard Deviation (σ):

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

Sample Standard Deviation (s):

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

- Variance is in **squared units** (e.g., cm^2, $marks^2$).
- Standard deviation brings it back to the **original units** (e.g., cm, marks).
- If SD is **small** → data is tightly clustered around the mean.
- If SD is **large** → data is more spread out.

Measuring the Dispersion of Data

Example (using our previous data: 2, 4, 4, 4, 6)

We already found:
Population variance = **1.6**
Sample variance = **2**

So:

Population SD (σ):

$$\sigma = \sqrt{1.6} \approx 1.26$$

Sample SD (s):

$$s = \sqrt{2} \approx 1.41$$

Measuring the Dispersion of Data

- Variance is in **squared units** (e.g., cm^2, $marks^2$).

- Standard deviation brings it back to the **original units** (e.g., cm, marks).

- If SD is **small** → data is tightly clustered around the mean.
- If SD is **large** → data is more spread out.

Measuring the Dispersion of Data

Scores = [75, 81, 95, 85, 81, 82, 97, 89, 76, 100, 77, 79, 87, 87, 79, 75, 100, 82, 83, 98, 86, 87, 71, 78, 81]

Data facts

- $n = 25$
- $\sum x_i = 2111$
- Mean $\bar{x} = \frac{2111}{25} = 84.44$ (exact)

Squared-deviation sum

$$\sum (x_i - \bar{x})^2 = 1616.16$$

Population (divide by n)

$$\sigma^2 = \frac{1616.16}{25} = 64.6464 \ \ (\approx 64.65), \quad \sigma = \sqrt{64.6464} \approx 8.04$$

Sample (divide by $n - 1$)

$$s^2 = \frac{1616.16}{24} = 67.34, \quad s = \sqrt{67.34} \approx 8.21$$

Measuring the Dispersion of Data

- Quartiles, outliers and boxplots
  - Quartiles: Q_1 (25th percentile), Q_3 (75th percentile)

- **Q1 (First Quartile / 25th Percentile):**
  25% of the data lies **below** Q1, and 75% lies **above** it.
- **Q3 (Third Quartile / 75th Percentile):**
  75% of the data lies **below** Q3, and 25% lies **above** it.

  - Inter-quartile range: IQR = $Q_3 - Q_1$
  - Five number summary: min, Q_1, Median, Q_3, max
  - Outlier: usually, a value higher/lower than 1.5 x IQR

Measuring the Dispersion of Data

Data (sorted with positions)

1: 71, 2: 75, 3: 75, 4: 76, 5: 77, 6: 78, 7: 79, 8: 79, 9: 81, 10: 81, 11: 81, 12: 82, 13: 82, 14: 83, 15: 85, 16: 86, 17: 87, 18: 87, 19: 87, 20: 89, 21: 95, 22: 97, 23: 98, 24: 100, 25: 100

n = 25

Median (Q2)
For odd n, position = (n + 1) / 2 = (25 + 1)/2 = 13 → value at position 13 = **82**.
Median (Q2) = 82

Measuring the Dispersion of Data

General index: $h = 1 + (n - 1)\,p$ where p is the percentile in decimal.

- **Q1 (25th percentile):**

$$h_1 = 1 + (25 - 1) \times 0.25 = 1 + 24 \times 0.25 = 7$$

- Take value at position 7 → **Q1 = 79**

- **Q3 (75th percentile):**

$$h_3 = 1 + (25 - 1) \times 0.75 = 1 + 24 \times 0.75 = 19$$

Take value at position 19 → **Q3 = 87**

Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1 = 87 - 79 = \mathbf{8}$$

Measuring the Dispersion of Data

Outlier Bounds (1.5 × IQR rule)

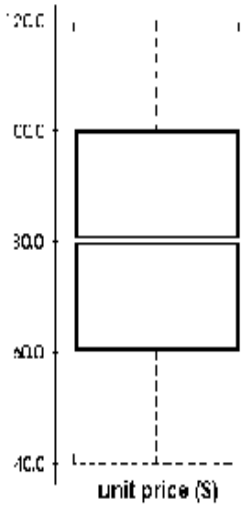
- **Lower bound** = $Q1 - 1.5 \times IQR = 79 - 1.5 \times 8 = 79 - 12 = \mathbf{67}$
- **Upper bound** = $Q3 + 1.5 \times IQR = 87 + 1.5 \times 8 = 87 + 12 = \mathbf{99}$

Any value < 67 or > 99 is an outlier.

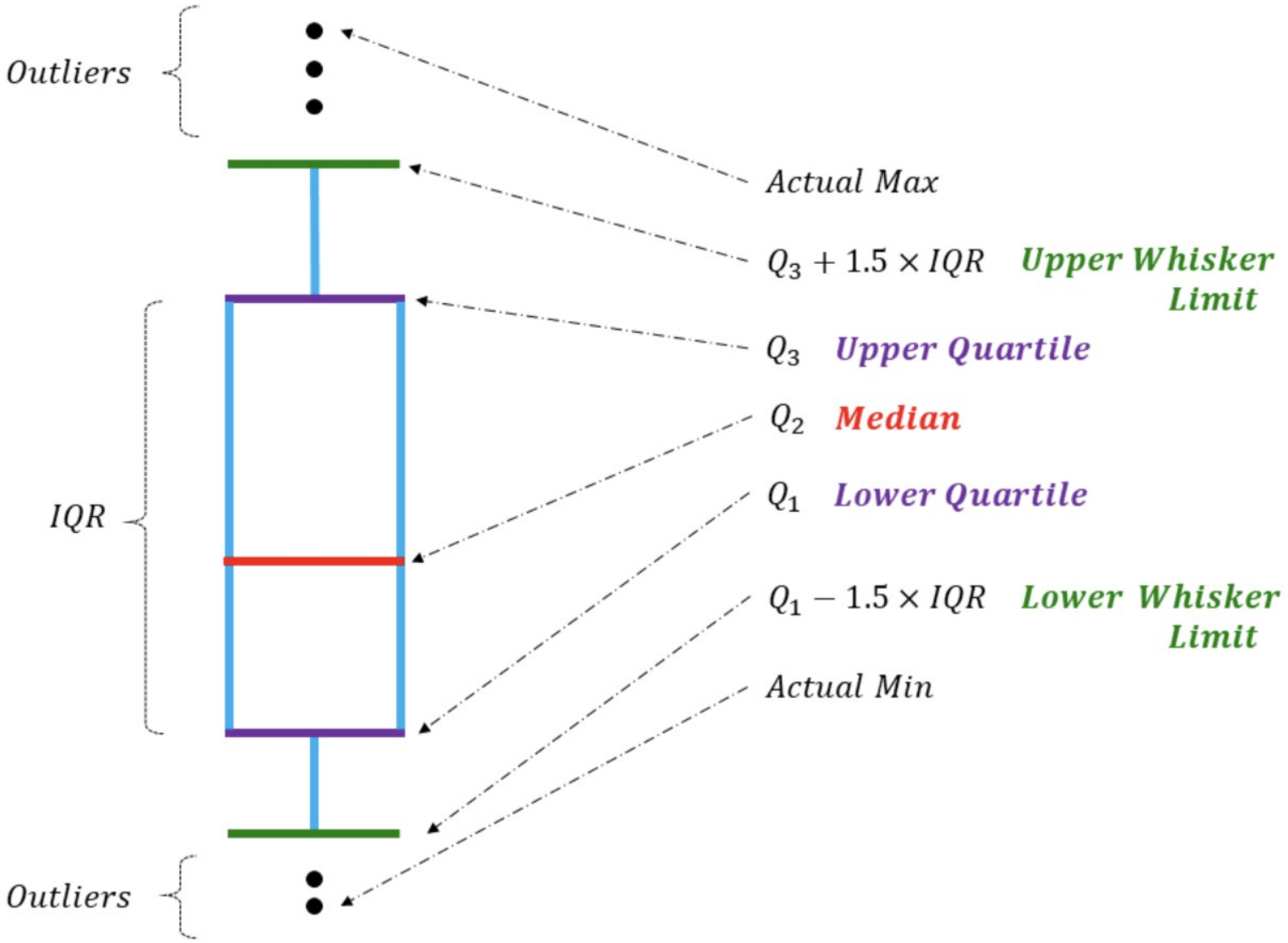
Outliers: 100, 100 (both > 99)

Boxplot Analysis

- Five-number summary of a distribution:
  - Minimum,
  - Q1,
  - M,
  - Q3,
  - Maximum



Boxplot Analysis



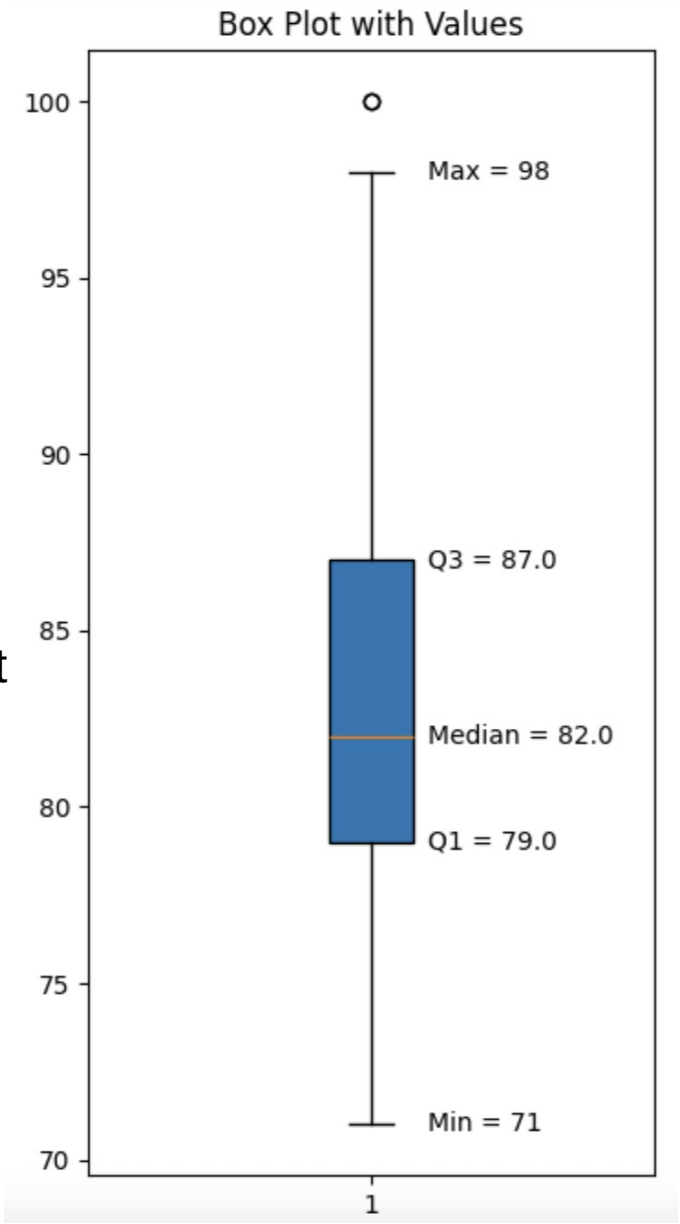
BOXPLOT ANALYSIS

How Boxplot Whiskers Are Determined

- **Q1 = 79**
- **Median (Q2)= 82**
- **Q3 = 87**
- **IQR = Q3 – Q1 = 8**
- **Upper whisker limit = 99**
- **Lower whisker limit = 67**

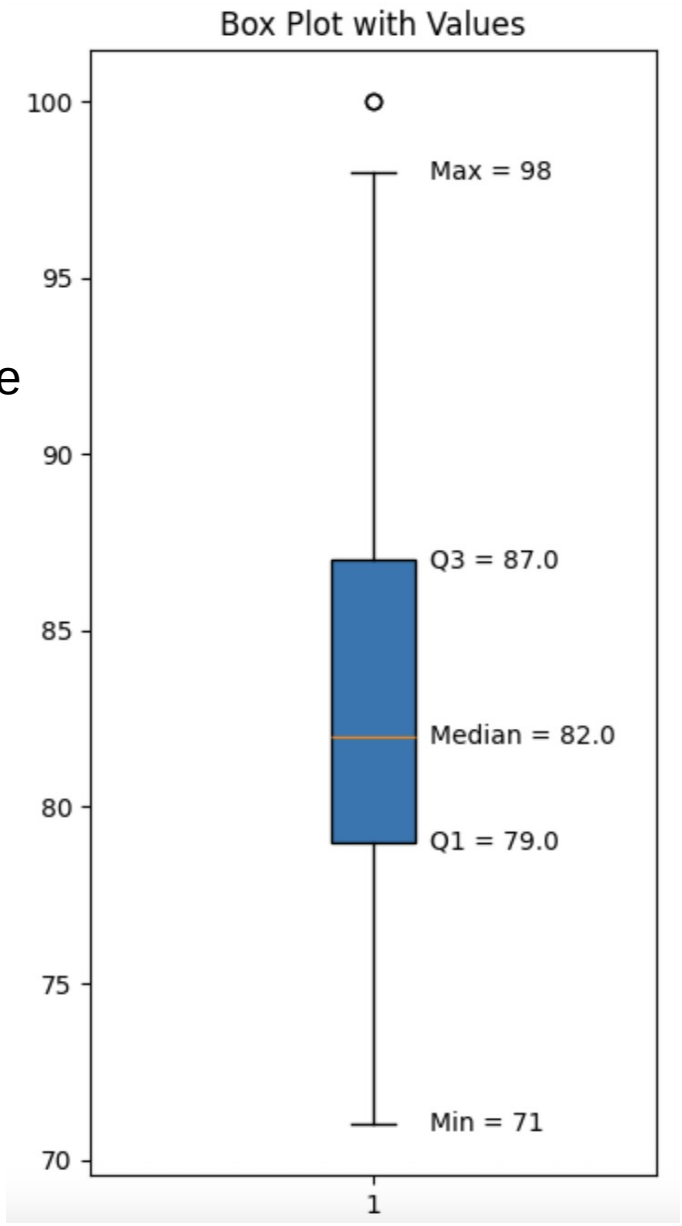
- Any value ≤ 99 is included in the **upper whisker**.
- Any value > 99 is marked as an **outlier (dot)**.

Lower whisker = the **smallest actual data point** that is **≥ lower fence (67)**.
Our data's minimum = 71
Since 71 ≥ 67, the lower whisker is drawn at **71**.



BOXPLOT ANALYSIS

- Largest value = 100
- Since 100 > 99 → Matplotlib **does not include it in the whisker**
- Instead, the whisker stops at the largest value ≤ 99 (which is **98**)
- The **100s appear as outliers** (the little circles above the whisker)



Comparing Distributions

- Are the two distributions similar or dissimilar?
- Often there is a need to compare different distributions to derive some important insights or make decisions.
- Visual inspection via comparing histograms (or any graphical depiction of distribution) is not practical (and not easy).
- Visual inspection via comparing box plots.

Hypothesis testing and **Confidence or probability value** are used for comparing distributions.

What is Hypothesis Testing?

Hypothesis testing is a **statistical procedure** to decide, using sample data, whether to accept or reject a certain belief (assumption) about a population.

Key Terms

Null Hypothesis (H_0):

- The default belief or "status quo".
- Example: *"There is no difference between the two group means."*

Alternative Hypothesis (H_1 or Ha):

- The opposite of H_0, what we want to prove.
- Example: *"There is a difference between the two group means."*

Test Statistic:

- A number calculated from sample data (e.g., t-statistic, z-statistic).
- Used to decide whether data is consistent with H_0.

P-Value

- *p*-value is the confidence or probability value of the null hypothesis.
- It indicates how much we believe the two distributions are the same.
- If *p*-value is very small, then we can reject the null hypothesis, and accept the alternative hypothesis.
- If *p*-value is not very small, then we accept the null hypothesis.

Significance Level (α):

- A threshold set before the test (commonly $\alpha = 0.05$).
- If **p-value < α → reject H_0**.

If **p < 0.05** → reject null hypothesis

If **p ≥ 0.05** → fail to reject null hypothesis

Comparing Distributions

Typical Tests

- Shapiro-Wilk Test: Test if a data sample follows **normal distribution.**
  - Null hypothesis: the sample has a normal distribution.
- Student's t-test: Tests if the **means** of two independent samples are significantly different. **Assume normal distributions.**
  - Null hypothesis: the means of the samples are equal.
- Mann-Whitney U Test: Tests if of two distributions are significantly **different or similar**. Not normal distribution.
  - Null hypothesis: the underlying distributions are similar